Global Renewable Energy in 2025: How Far Have We Come?

Global trends across primary energy and electricity • Source: Our World in Data

A data-driven overview of renewable energy’s growing share in the global energy and elec

Audience: Policy analysts & energy sector strategists

75% of Global GHG Emissions Still Come from Fossil Fuels

~75%
of global GHG emissions from burning fossil fuels

5M+
premature deaths per year from air pollution

Core task
shift to low-carbon energy (renewables + nuclear)

Why this matters

Fossil fuels have dominated energy systems since the Industrial Revolution.
Decarbonization requires rapidly scaling low-carbon electricity and extending it into transport and heating.
Modern renewables (solar, wind, hydropower, geothermal, bioenergy, wave, tidal) are central levers; traditional biomass is excluded from modern renewable statistics.

Renewables Now Supply ~14% of Global Primary Energy

2024 (substitution method) • Primary energy = electricity + transport + heating

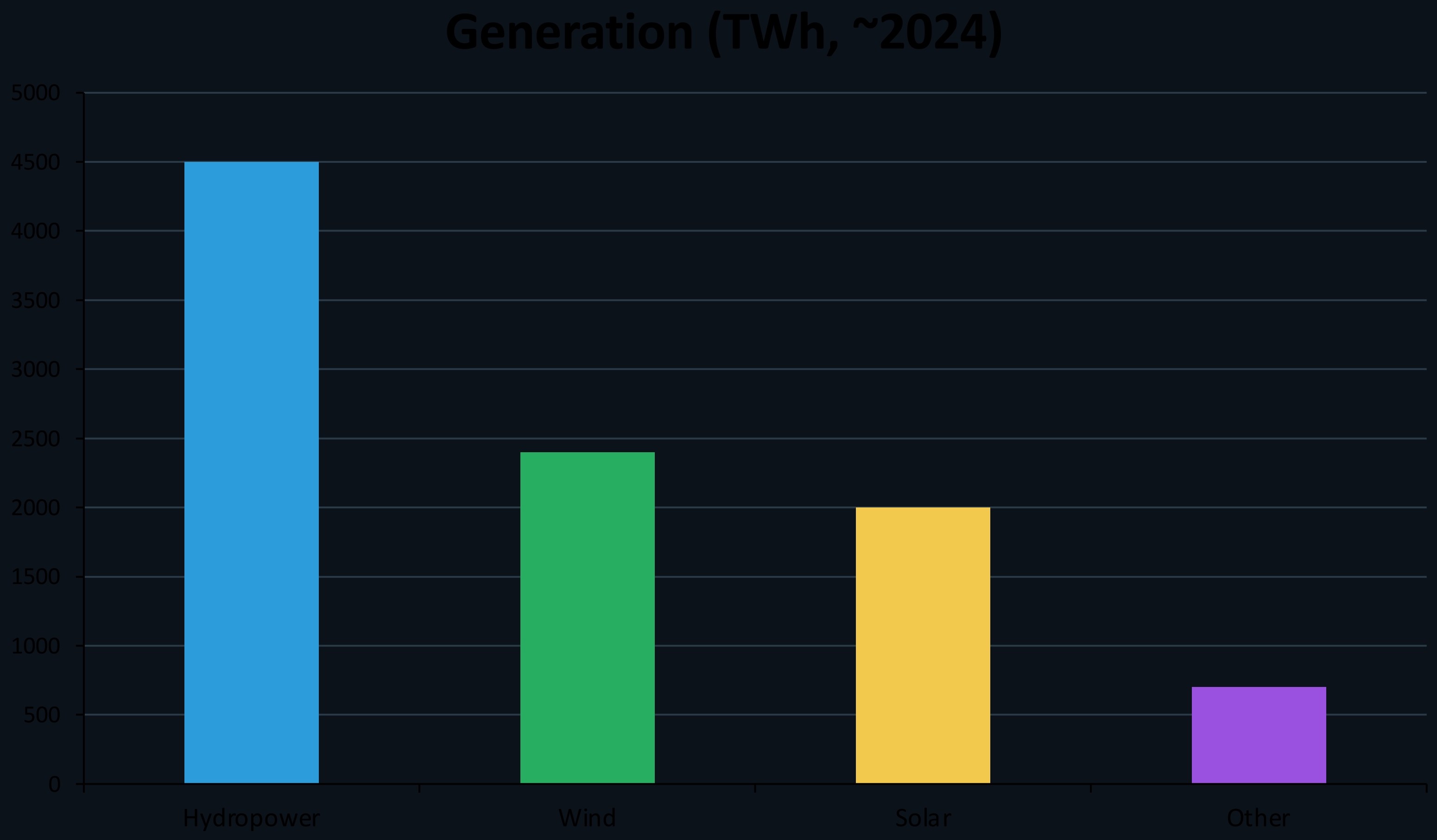
~14%

global primary energy from renewables

Why lower than electricity?

Transport and heating remain more reliant on oil and gas, keeping renewables’ share of total energy lower than in electricity alone.

Renewable Electricity Generation Is Growing Rapidly, Led by Wir

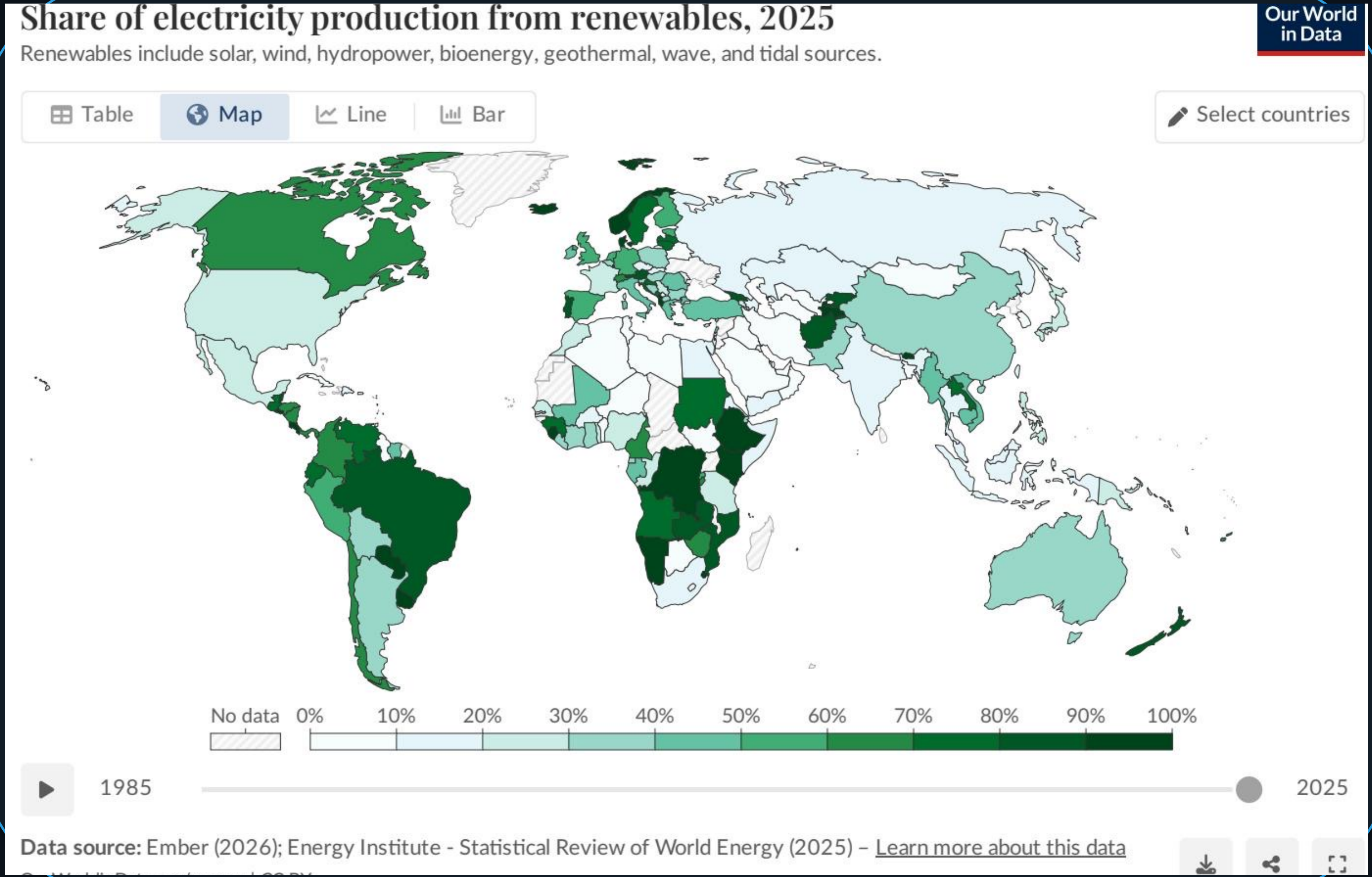


What the breakdown shows

Hydropower is still the largest contributor (~4,500 TWh).
Wind (~2,400 TWh) and solar (~2,000 TWh) are now close to hydro when combined.
Wind and solar acceleration since ~2010 is the key driver of recent growth.

Almost One-Third of Global Electricity Now Comes from Renewa

2025 • Large cross-country variation



Selected shares (approx.)

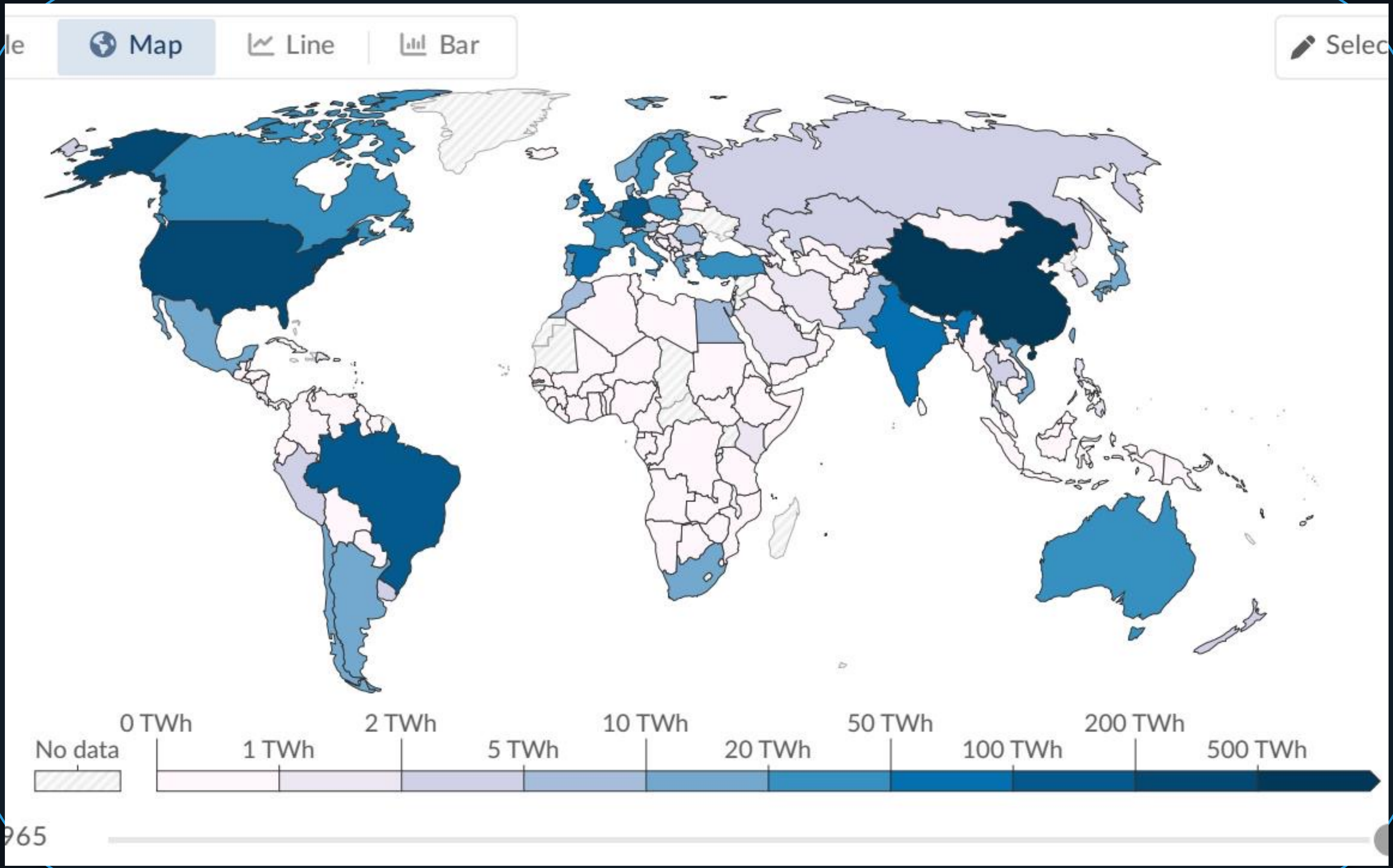
Country/Region	Renewables
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Global average: ~30% renewable electricity

Leaders reach 60–90%+; many Middle Eastern and Afric

Hydropower Remains the Largest Renewable Source at ~50% of

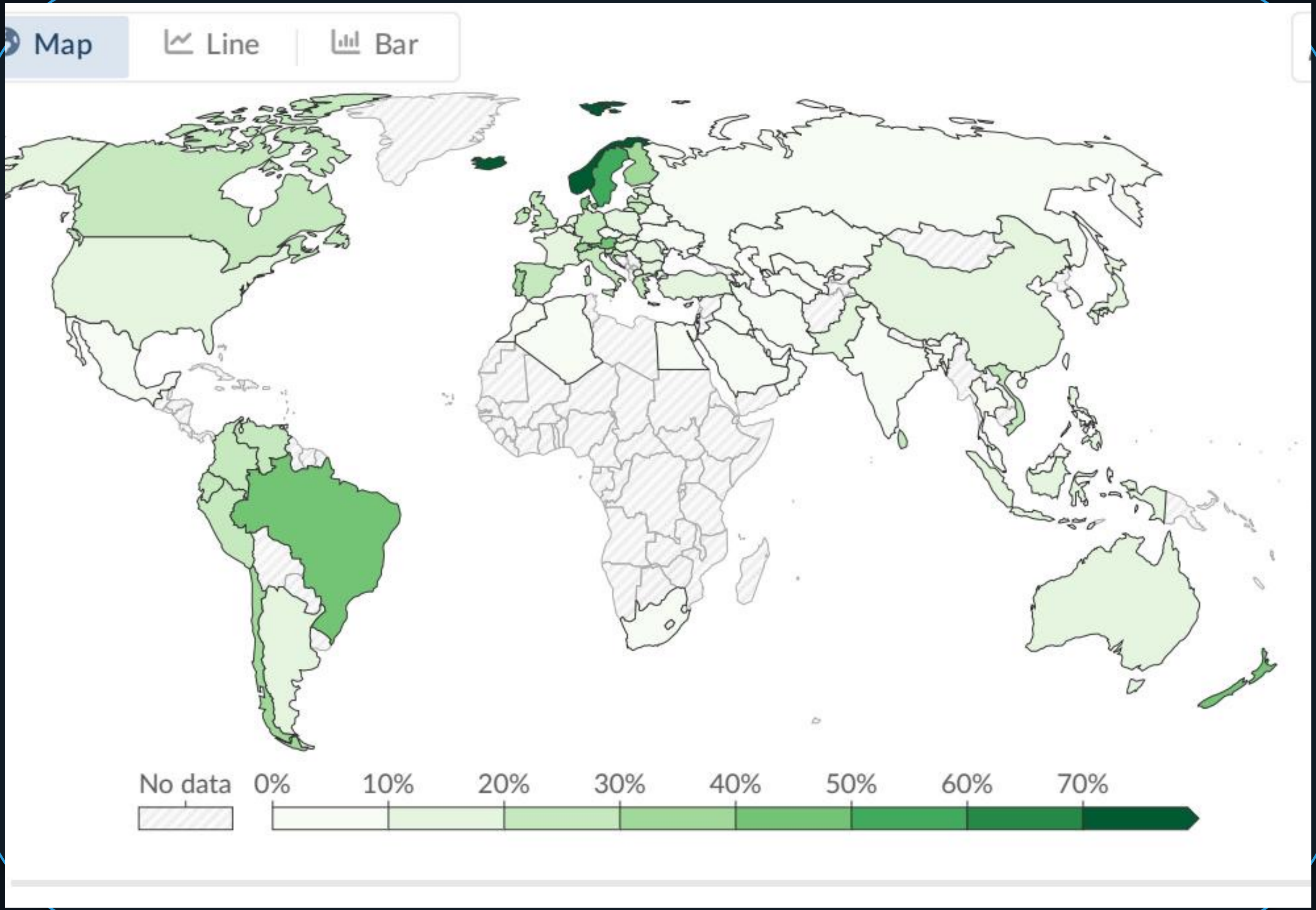
Hydro ≈ ~47% of renewable electricity (~4,500 TWh)



Policy/strategy notes

Largest renewable electricity source (~half).
Top producers: China, Brazil, Canada.
Mature technology with site constraints; expansion potential is more limited than wind/solar in many regions.

Wind and Solar Are the Fastest-Growing Energy Technologies in



Technology	TWh	Share
Wind	~2,400	~25%
Solar	~2,000	~21%

Wind generation

Wind vs. So

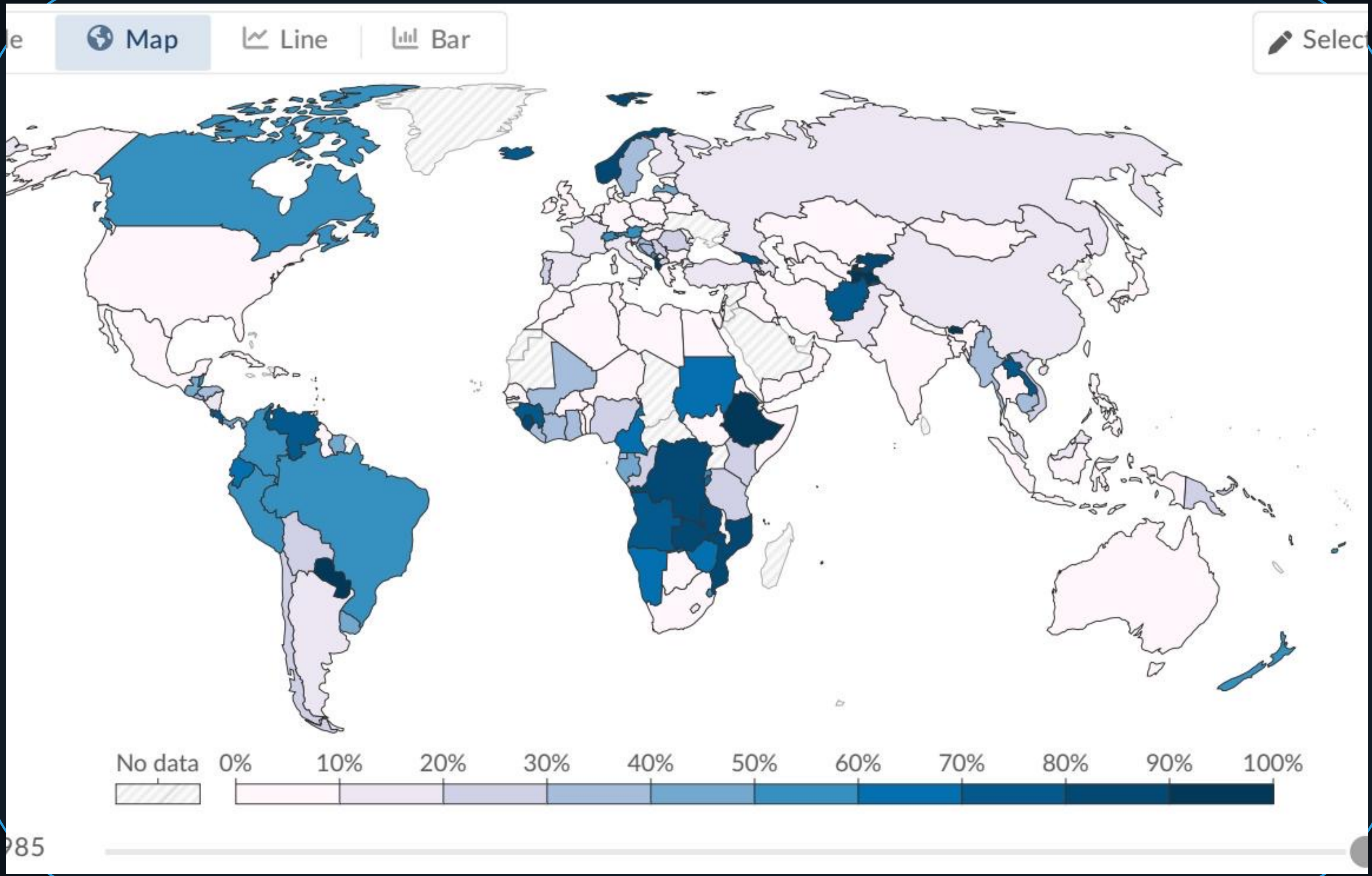
Solar is highlighted as the fastest-growing source since 2010, driven by falling costs and policy support.

Wind + solar combined r

Solar generation

output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia



Uneven transition dynamics

Scandinavia, Brazil, and Canada benefit from abundant hydropower (often 65–98% renewable electricity).

Denmark and Germany have invested heavily in wind and solar.

Many Sub-Saharan African countries have minimal renewable electricity infrastructure.

Many Middle Eastern nations remain almost entirely fossil-fuel dependent, often below 10% renewable electricity.

Key Takeaways: Renewables Are Growing Fast but the Transition

What to remember

Renewables are ~14% of global primary energy (2024, substitution method) — lower than electricity because transport/heating lag.
Renewables are ~30% of global electricity (2025), with countries ranging from <2% to ~98%.
Hydropower is still dominant (~47% of renewable electricity), but wind (~25%) and solar (~21%) are catching up fast.
Regional disparities are stark: Scandinavia and Latin America lead; much of the Middle East and Sub-Saharan Africa remains below 10% renewable electricity.
Strategic implication: accelerate deployment and grid integration now, while expanding electrification beyond power into transport and heating.

Forward look: keep scaling wind and solar, modernize grids, and electrify end uses to raise renewables' share beyond electricity.